LECTURE 2

GEOSPATIAL DATA: IMAGES, MAPS, TIME SERIES

Geospatial Representation Learning

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Learning Outcomes

Lecture

- Identify major geospatial data types, including satellite imagery, EO time series, vector GIS layers, maps, and multimodal datasets.
- Compare spatial, temporal, spectral, and semantic properties of different geospatial data sources.
- Explain preprocessing steps such as reprojection, resampling, tiling, masking, and temporal aggregation.
- Assess which data sources are suitable for different geospatial machine learning tasks.

Lab

- Access and preprocess raster, vector, and time-series geospatial data using Python-based tools.
- Combine multiple geospatial data layers into a common spatial reference and resolution.
- Extract training samples or analysis regions from geospatial datasets.
- Evaluate practical limitations of data quality, coverage, scale, and interoperability.

PRACTICAL 2

WORKING WITH GEOSPATIAL DATA SOURCES

Geospatial Representation Learning

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